

20 General References

Ferro Corporation. Technical literature: *Synpro Aluminum Monostearate NF*, 2008.
 Mallinckrodt. Technical literature: *HyQual Aluminum stearate*, 2008.

21 Author

J Shur.

22 Date of Revision

19 February 2009.



Aluminum Oxide

1 Nonproprietary Names

None adopted.

2 Synonyms

Activated alumina; activated aluminum oxide; alpha aluminum oxide; alumina; alumina, calcined; alumina, tabular; aluminum oxide alumite; aluminum trioxide; gamma aluminum oxide.

3 Chemical Name and CAS Registry Number

Aluminum oxide [1344-28-1]

4 Empirical Formula and Molecular Weight

Al_2O_3 101.96

5 Structural Formula

Aluminum oxide occurs naturally as the minerals bauxite, bayerite, boehmite, corundum, diaspore, and gibbsite.

6 Functional Category

Adsorbent; dispersing agent.

7 Applications in Pharmaceutical Formulation or Technology

Aluminum oxide is used mainly in tablet formulations.⁽¹⁾ It is used for decoloring powders and is particularly widely used in antibiotic formulations. It is also used in suppositories, pessaries, and urethral inserts. Hydrated aluminum oxide (see Section 18) is used in mordant dyeing to make lake pigments, in cosmetics, and therapeutically as an antacid.

8 Description

Aluminum oxide occurs as a white crystalline powder. Aluminum oxide occurs as two crystalline forms: α -aluminum oxide is composed of colorless hexagonal crystals, and γ -aluminum oxide is composed of minute colorless cubic crystals that are transformed to the α -form at high temperatures.

9 Pharmacopeial Specifications

See Section 18.

10 Typical Properties

Boiling point 2977°C
Density (bulk) 0.9–1.1 g/cm³
Flammability Nonflammable.
Hardness (Mohs) 8.8
Hygroscopicity Very hygroscopic.

Melting point 2050°C

Solubility Slowly soluble in aqueous alkaline solutions with the formation of hydroxides; practically insoluble in nonpolar organic solvents, diethyl ether, ethanol (95%), and water.

Specific gravity 2.8 (becomes 4.0 at 800°C)

Vapor pressure 133.3 Pa at 2158°C

11 Stability and Storage Conditions

Aluminum oxide should be stored in a well-closed container in a cool, dry, place. It is very hygroscopic.

12 Incompatibilities

Aluminum oxide should be kept well away from water. It is incompatible with strong oxidizers and chlorinated rubber. Aluminum oxide also reacts with chlorine trifluoride, ethylene oxide, sodium nitrate, and vinyl acetate. Exothermic reactions above 200°C with halocarbon vapors produce toxic hydrogen chloride and phosgene fumes.

13 Method of Manufacture

Most of the aluminum oxide produced commercially is obtained by the calcination of aluminum hydroxide.

14 Safety

Aluminum oxide is generally regarded as relatively nontoxic and nonirritant when used as an excipient. Inhalation of finely divided particles may cause lung damage (Shaver's disease).⁽²⁾

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of the material handled.⁽³⁾ In the UK, the workplace exposure limits for aluminum oxide are 10 mg/m³ long-term (8-hour TWA) for total inhalable dust and 4 mg/m³ for respirable dust.⁽⁴⁾ In the USA, the OSHA limit is 15 mg/m³ total dust, 5 mg/m³ respirable fraction for aluminium oxide.⁽⁵⁾

16 Regulatory Status

Included in the FDA Inactive Ingredients Database (oral tablets and topical sponge). Included in nonparenteral medicines licensed in the UK.

17 Related Substances

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18 Comments

A specification for aluminum oxide is included in the *Japanese Pharmaceutical Excipients* (JPE);⁽⁶⁾ see Table I. A specification for

light aluminum oxide is also included. The PhEur 6.3 includes a specification for hydrated aluminum oxide that contains the equivalent of 47.0–60.0% of Al_2O_3 .

The EINECS number for aluminum oxide is 215-691-6.

Table I: JPE specification for aluminum oxide.⁽⁶⁾

Test	JPE 2004
Identification	+
Water-soluble substances	+
Heavy metals	≤30 ppm
Lead	≤30 ppm
Arsenic	≤5 ppm
Loss on drying	≤1.5%
Loss on ignition	≤2.5%
Assay	≥96.0%

19 Specific References

- 1 Rupperecht H. Processing of potent substances with inorganic supports by imbedding and coating. *Acta Pharm Technol* 1980; 26: 13–27.

- 2 Lewis RJ, ed. *Sax's dangerous Properties of Industrial Materials*, 11th edn. New York: Wiley, 2004; 136.
- 3 National Poisons Information Service 1997. Aluminium oxide. <http://www.intox.org/databank/documents/chemical/alumoxide/ukpid33.htm> (accessed 16 January 2009)
- 4 Health and Safety Executive. *EH40/2005: Workplace Exposure Limits*. Sudbury: HSE Books, 2005 (updated 2007). <http://www.hse.gov.uk/coshh/table1.pdf> (accessed 5 February 2009).
- 5 JT Baker (2007). Material safety data sheet: Aluminium oxide. <http://www.jtbaker.com/msds/englishhtml/a2844.htm> (accessed 5 February 2009).
- 6 Japan Pharmaceutical Excipients Council. *Japanese Pharmaceutical Excipients 2004*. Tokyo: Yakuji Nippo, 2004; 67–68.

20 General References

21 Author

T Farrell.

22 Date of Revision

5 February 2009.

Aluminum Phosphate Adjuvant

1 Nonproprietary Names

None adopted.

2 Synonyms

Adju-Phos; aluminum hydroxyphosphate; aluminium hydroxyphosphate; *Rehydraphos*.

3 Chemical Name and CAS Registry Number

Aluminum phosphate [7784-30-7]

4 Empirical Formula and Molecular Weight

$\text{Al}(\text{OH})_x(\text{PO}_4)_y$

The molecular weight is dependent on the degree of substitution of phosphate groups for hydroxyl groups.

5 Structural Formula

Aluminum phosphate adjuvant occurs as a precipitate of amorphous aluminum hydroxide in which some sites contain phosphate groups instead of hydroxyl. Both hydroxyl and phosphate groups are exposed at the surface. The hydroxyl groups produce a pH-dependent surface charge by accepting a proton to produce a positive site, or donating a proton to produce a negative site. The pH-dependent surface charge is characterized by the point of zero charge, which is equivalent to the isoelectric point in protein chemistry. The surface hydroxyl groups may also undergo ligand exchange with fluoride, phosphate, carbonate, sulfate, or borate anions.

Aluminum phosphate adjuvant is not a stoichiometric compound. Rather, the degree of phosphate group substitution for hydroxyl groups depends on the precipitation recipe and conditions.

6 Functional Category

Adsorbent; vaccine adjuvant.

7 Applications in Pharmaceutical Formulation or Technology

Aluminum phosphate adjuvant is used in parenteral human and veterinary vaccines.⁽¹⁾ It activates Th2 immune responses, including IgG and IgE antibody responses.

8 Description

Aluminum phosphate adjuvant is a white hydrogel that sediments slowly and forms a clear supernatant.

9 Pharmacopeial Specifications

10 Typical Properties

Acidity/alkalinity pH = 6.0–8.0

Al:P atomic ratio 1.0–1.4:1.0

Aluminum (%) 0.5–0.75

Particle size distribution Primary particles are platy with an average diameter of 50 nm. The primary particles form aggregates of 1–10 μm .

Point of zero charge pH = 4.6–5.6, depending on the Al:P atomic ratio.

Protein binding capacity >0.6 mg lysozyme/mg equivalent Al_2O_3

Solubility Soluble in mineral acids and alkali hydroxides.

X-ray diffractogram Amorphous to x-rays.

11 Stability and Storage Conditions

Aluminum phosphate adjuvant is stable for at least 2 years when stored at 4–30°C in well-sealed inert containers. It must not be